

Homework 2

CFRM 505

Due Sunday, April 19 at 11:59pm

1 Problem 1

Use the inverse transform method to estimate the mean and variance of the following random variables (using at least 10,000 samples for each part):

1.1 Part a

$X = \min \{X_1, X_2, X_3\}$ where X_1, X_2 and X_3 are i.i.d. exponentially distributed random variables with parameter $\lambda = 2$.

1.2 Part b

Y with the pdf

$$f_Y(y) = \begin{cases} \frac{3}{8}y^2e^{-y^3/8} & y > 0 \\ 0 & \text{otherwise} \end{cases}$$

2 Problem 2

Use Monte Carlo simulation to estimate the mean and variance of $X \sim \text{Poisson}(10)$. Plot a histogram from your data. Use the inverse transform method to generate X and use at least 10,000 samples.

3 Problem 3

Consider the random variable X whose cdf is given by

$$F_X(x) = \int_0^\infty 2x^ye^{-2y} dy \text{ for all } 0 < x < 1$$

Use Monte Carlo simulation to estimate the mean and variance of this distribution. Use the following methods to generate samples of X (using at least 10,000 samples for each part).

3.1 Part a

Find a closed form expression for F_X (i.e., do the integral) and then use the inverse transform method.

3.2 Part b

Use the composition method. (Hint: Use $f_Y(y) = 2e^{-2y}$ and $F_{X|Y}(x; y) = x^y$.)

4 Problem 4

Consider the following “game”:

- 1) Set $r = 1$ and $n = 1$
- 2) Generate a random vector (X, Y) uniformly distributed on the square $[-1, 1] \times [-1, 1]$.
- 3) If $X^2 + Y^2 > r^2$, then stop.
- 4) If $X^2 + Y^2 \leq r^2$, then set r equal to $\sqrt{r^2 - X^2 - Y^2}$ and set n equal to $n + 1$, then start over at step (2).

Your score at the end of this game is $N = n$ (the number of random vectors you calculated). Use Monte Carlo simulation with at least 10,000 samples to simulate this game and estimate the expected score $\mathbb{E}[N]$.

(This game is taken from [the lesson by Grant Sanderson in 3b1b](#). The link has a lovely proof that the expected value is actually $e^{\pi/4}$.)

5 Problem 5

Let $\mathbf{X} = (X, Y)$ be uniformly distributed over the unit circle.

5.1 Part a

Find the marginal pdf $f_Y(y)$ and the conditional pdf $f_{X|Y}(x; y)$.

5.2 Part b

Simulate 1,000 samples of (X, Y) using the composition method. Plot your samples on the xy -plane to confirm that they are uniformly distributed. (You are welcome to use the plotting code from section 4.10 of the notes, currently on page 38. You will have to write a new version of `generate_point()` using the composition method.)

Hint: The marginal pdf $f_Y(y)$ is quite simple, but the marginal cdf $F_Y(y)$ is complicated enough that you can't invert it by hand. I recommend that you still use the inverse cdf method, but invert the cdf numerically. In particular, to generate a value of Y you will need to solve the equation $F_Y(y) = U$. You can do so using the python function `scipy.optimize.root_scalar()`. In particular, the code `y = scipy.optimize.root_scalar(lambda y: f(y), bracket=(a, b)).root` finds a solution to the equation $f(y) = 0$ with $a \leq y \leq b$.